4-3 Milestone: Hash Table Data Structure Pseudocode

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Initialize a hash table data structure to store course information.

Function loadCoursesFromFile(filename):

Open the file with the given filename.

If the file cannot be opened, print an error message and exit.

Initialize an empty list to store prerequisites temporarily.

Read each line from the file:

Split the line into tokens using a delimiter (e.g., comma or tab).

If the number of tokens is less than 2, print an error message for invalid file format and skip to the next line.

Extract courseNumber, title, and prerequisites from the tokens.

If there are prerequisites:

Split the prerequisites string into individual prerequisite course numbers.

For each prerequisite course number:

Check if the prerequisite course exists in the hash table:

If it does not exist, print an error message and skip to the next line.

Otherwise, add the prerequisite course number to the temporary list.

Create a course object with courseNumber, title, and prerequisites list.

Insert the course object into the hash table using courseNumber as the key.

Close the file.

Function printCourseInformation():

Iterate over each entry (courseNumber, courseObject) in the hash table:

Print courseNumber, title, and prerequisites of the courseObject.